

ActInf Livestream #005.0: Context for "Multiscale integration: beyond internalism..." (2019)

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hello and welcome to the active inference live stream my name is daniel friedman and this is going to be a solo discussion if you're a new listener or not welcome to team com we are an experiment in online team communication and learning related to active inference you can find us on twitter at inferenceactive at activeinference gmail.com on key base via our public team or our youtube channel this is a recorded and archived live stream so please provide us with feedback so that we can improve our work all backgrounds and perspectives are welcome here and also as far as video etiquette goes remember to mute if there's noise in the background raise your hand if you have a question and so on here we are in act imp stream number 5.0 we're kind of exploring with the numbering system so give us any feedback as far as what kinds of numbers or what kinds of shows might be helpful for your learning but the goal for this discussion is to set the context for active in stream 5.1 in 5.1 we're going to discuss the paper multi-scale integration beyond internalism and externalism by ramstead at all 2019. the goal of this video is really to contextualize some of the ideas the historical threads and the vocabulary that will help you either read or work with the ramstead paper uh in terms of other fields that you might be more familiar with or just to sort of connect ideas and thus ground your understanding and your practice so i come from a different background from the author so i have a slightly different perspective on perhaps even the implications of the paper so the goal here is really just to explore that so that whether you listen to this by itself and you're looking to sort of see where active inference is trying to relate itself to other philosophical ideas or whether you're coming from the active inference side and you want to understand some of the broader philosophical questions that are being addressed potentially by active inference in both those cases i hope this is an interesting video for you so here are a few of the topics that we'll be exploring here the first one is top-down and bottom-up causation and that's going to be an introduction to multi-level systems as well as thinking about causal modeling also next we'll talk about and raise some questions related to partitioning between

internal and external cognition which is certainly the wedge or the main issue that the ramps that at all paper is going to be focused on and then we'll talk about a few different perspectives on systems modeling and action that's sort of action and inference if you will but there's also been a lot of philosophical thought on systems modeling as well as action so we can kind of see how those two fields might come together in the field called active inference and next time in number 5.1 we will all come together everybody who wants to can always get in touch with us we will all come together and discuss the paper to whatever extent we've read it whatever we thought about it any questions that we wanted to address any of our background knowledge uh any of our perspectives that we bring to the paper those are all cool things to come about in 5.1 and if you're listening to this after 5.1 has happened you can either listen to 5.1 or you could get in touch with us to be on a future discussion so next time in 5.1 we're going to be talking about the figures to abstract etc in this paper by ramstead at all called multi-scale integration beyond internalism and externalism and that's in the journal syntheses in 2019 so the goal of that paper and the only quotes from the paper that are going to be in this video are just from the introduction and abstract just so that we can ground the ideas more specifically in what ramps that at all frame their work as so the goal of the paper in the author's words was we focus on making explicit a description of the boundaries of cognitive systems that we think follows from taking seriously the inactive embodied and extended nature of cognition this is the idea that the boundaries of cognitive systems are nested and multiple and that with respect to its study cognition has no fixed or essential boundaries cool i think that that is really exciting and also it doesn't necessarily bring the free energy principle or active inference on too heavily it just helps understand what is the broader goal of the paper well the question that really excited me today was this taking it seriously aspect on it's the first two words of the paper's title so maybe they're trying to signal with that the story that we want to be working towards is multi-scale integration so linking things that are different spatial and temporal scales at once multi-scale how are we going to do that multi-scale integration well what they're provocatively stating in the title is that we must move beyond this debate of internalism versus externalism which i'll contextualize soon and the way that they're going to take it seriously is using a mathematical framework called free energy principle and it's related process theory active inference so that's what taking it seriously looks like for them especially drawing on their background as cognitive science experts they're really framing this as about multi-scale cognition and i came from a not a cognitive sciences background more of an evolutionary biology background and so multi-scale integration in the ant colony or in the

insect brain or in the cell there's a different set of philosophical baggage that comes along with it sometimes if i could characterize it so clearly would be the focus on humans tends to focus on the experience and the internal view since we are human so at the very least it's one of the ways that we can study emergencies by ourself as an emergent process whereas in computers and colonies the cell there's often a tendency following let's just say dennit and others to imagine it as an automata or just sort of a robot and therefore devoid of experience or specularity sorry speculatively having experience but sort of on the sci-fi fringe and anyways when you deal with the system you have to be in that intentional stance so those types of systems tend to have perspectives that are devoid of agency whereas human systems the debate is just about that is about our own agency but we're all part of the same multi-skill integrated framework and so we want to have a story that works for cells that are in our gut that are in the uh in our gut lumen in the microbiome why should it matter why should we have a different theory of biology for one side of that over the other especially when we know that there's important transfer between them for example so the authors took it seriously and they wrote this 2019 paper as well as other papers i'm gonna try to take this seriously from the work in biology that i have learned about over the years and that's what taking it seriously and contextualizing looks like to me so i just thought i could start with that here's a paper from 2012 by dennis noble who's a great systems biologist called a theory of biological relativity no privileged level of causation so you can kind of see where he's going which is taking the idea of relativity whether that's gravitational relativity something related to quantum mechanics for example or just uh inter-subjective relativity like focusing on the relations that define something or the interactions that define a person rather than an essentialist take so noble is sort of taking a broad understanding of relativity and saying well if we really look at multi-scale systems and spatial and temporal cause then there's going to be this relativity principle maybe that could apply to biology and what i like about his figure one and i've added two wrongs at the top is that it really clearly phrases what he's arguing against and also makes it clear um how some of these logical errors become deployed in real systems in real analyses so figure one is a model of causation for what's called upward causation and the way to read this figure is it starts at the bottom with the smallest thing which are genes which here is just i believe being meant as a string of dna but that's a whole other discussion let's just say that genes uh are able to cause proteins in rna which are another scale of organization bigger moving up towards protein and rna networks up through cells tissues organs organisms social interactions ecological technological governance informational systems and so on so this is a

sort of multi-scale systems perspective and the version of events that we often get is implicitly one of what noble is characterizing accurately as upward causation one could change the spatial metaphor it could be moving downwards it could be moving inwards or outwards but suffice to say that this is what we mean when we say upward causation is really from small things to big things because the top down things are often seen as big and the things that are just on the fabric level of reality are often seen as small rightly or wrongly um and the the story when you have this type of a upward causation model again that smaller things cause bigger things is the the genes are upstream of the behavior or um the uh the cell damage is upstream of organs so it makes sense and there's also a related idea called supervenience that is that is there a relationship between changing the smaller thing and the bigger thing changing so these are the kinds of questions that come up and this is one direction of causation however what noble ends up framing in this uh no privileged level of causation model is that yes there is influence or flow from so-called the upward direction so that's the genes and uh going all the way up in influencing things that are at larger scale but also in a very real and causal sense there are top-down causation that can be equally important for regulating the system so even if in the short time scale the in uh influence of the bottom up is extremely strong if there's no top-down force sometimes any amount of bottom-up power can just cause the system to dissipate and therefore not exist so in that sense all systems that we observe continuing to dissipate energy for example like active inference foragers informational foragers humans living creatures we can really start to see that there has to be some sort of a top-down organizing force otherwise it would be like a land uh slide or a lightning strike it would just dissipate away so he says look because the top-down forces are so important and over evolutionary time end up shaping the parameters and the set of what is possible or functional at the lower level it really needs to have an equal hand causally so we also need to look at this from a top-down bottom-up perspective lots of people have thought about these types of issues in systems theories and in other areas but the three areas that um i just want to emphasize here before bridging to active inference are that there's a continuum of larger and smaller things that has to basically be true because you can't have something bigger inside of something smaller and in the context of this debate that we're framing uh to get set up for which is between internalism and externalism you can think about things that are bigger as being more external and things that are smaller as being more internal so like the story about the smaller things is going into the organ into the cells into our cells but the story going out is about the ecosystem or the the constructed niche and these are things that are external and so that's

gonna come back later in the paper and then also um as a consequence of that there's this nestedness or this embeddedness to these different levels of analysis and there's again many vocabularies for this type of recursive or embedded system and hopefully we'll address some but also feel free to provide feedback these are just some of the general aspects of multi-scale systems and some of the questions about multi-scale systems that the paper is going to address eventually and the field of research but also there's some philosophical work from areas other than the inactive cognitive area all right so the next topic is going to be about partitioning between internal and external cognition which is another way of saying what is the internalism versus externalism debate about how are we going to move beyond that how might the free energy principle be involved in that and in a sense the question that this is asking is how do we do causal modeling of situations where internal and external states interact for example through sense and action so let's think about how sense and action are interacting even at one level of analysis what the ant perceives for example in its actions what you perceive and its actions and then immediately one of the questions that many people have asked is okay so how do we make that story multi-level because the ant we could simulate it as just a robot with inputs and outputs and sort of black box inside but it is also composed of the tissues and the cells and it seems a little bit ridiculous if we would have to simulate from the cell up to the ecosystem level so we have to be able to separate that model somewhere even if only for computational tractability or comprehensibility so we got to make this trade-off between making causal models of systems that are at one level of analysis but including things that are at a smaller level of analysis that can have consequences that bubble up for example a drug binding to a receptor can bubble up to behavior that's like a story about upward causation or someone's choice or someone's context could dictate their physiology through any number of mechanisms which food or molecules are available so there is going to be times where that cross-level interaction is really important for the stories that we tell for the narratives that how we understand the world um how we communicate about those things so how does the multi-level perspective relate to causal modeling if your story is about two people who are fighting is the story only going to be at that level or are you going to bring in society are you going to bring in the molecular stuff how much upper and lower context do you provide for causal modeling at a given level of analysis how is causal modeling related to the possible actions of a system also known as the adjacent possible for example what else could be and this just kind of reminds me of how when someone describes a near-miss or a situation that could have been another way they're not saying it could have been any any other way this thought experiment

has been carried out by Gould and others in the sense that yeah something happened a certain way which means it was permitted by reality and then you're envisioning a slightly different scenario for example the ball hits the other side of the roof and it ends up rolling down a different direction doing some other outcome and the ball still has the same physics and you're still the same person so it's not any alternate reality it's a very tightly constrained causal reality and what kinds of alternate realities do people propose or speculate about ones that are within the causal latent model of the world where all stories where you cross the freeway are dangerous for example so that's the causal model that underlies what we genuinely generally are talking about when we're talking about what could happen we're not saying anything could happen we're seeing a pretty tight model given about how we think about cause and effect in the system why is causal modeling so important to active inference and there's two roads into this I would say one perspective is why is causal modeling important from the point of view of the organism and another would be of the scientist or the observer so first from the point of view of the organism there's a lot of ways to frame exactly what it is an animal does and a really common area that people study this without the psycho baggage is in animal foraging so whether it's a rodent or an ant people or a bird people are really interested in the bioenergetics of and the decision-making that underlies how the animal uh goes about searching for food and how does that depend on the dynamics of the ecosystem and so one popular framework in studying foraging is optimal foraging theory which you can just by the name of it imagine is kind of like optimal investing strategy and there's a lot of variations on it but it's just saying what the ant is trying to do or what the bird is trying to do is given the costs and the benefits or as estimated by the bird the predicted costs and benefits of the environment and the processing time it's going to do this reward maximization and the good news about that approach is it often works a lot of times animals are near their predicted efficiency uh however other times it can be quite far off and it's not exactly apparent why that might be perhaps some constraint exists that was not modeled and so the reward maximization idea succeeds or it captures a story or an important variable about a foraging system as you might expect but it doesn't really go the full way in explaining how the model is embodied or carried out by the animal because it doesn't necessarily have the kind of computational processing that a human simulation might not to compare the brain or the niche to a computer but just to say that that even as a physical computer imagined is still something that's quite different than the models that are being used to study the error divergence by which ultimately people are going to be asking whether strategy was optimal in terms of its reward or not what active inference does is it says right the

overall reward model is intractable it contains variables that are inaccessible or just not realistic for the nematode to be calculating for example in any way directly but what the animal can be doing is engaging in this cycle between action and perception and so it is making these sensory hypotheses about the world embodied through action so for example the eyes are circading around very rapidly where are they moving they're moving to where your brain predict predicts that you need to resolve your uncertainty about your visual field most so that's why even though your actual visual resolution on your retina or the density of the cells the ability to perceive the fine granularity of vision is quite different across the back of your eye yet the whole visual field unless you have some sort of visual alteration is uniformly clear and doesn't have a blind spot so that's evidence that at least visually we're in a generative model of what's happening and it shouldn't be too far away to say that that's what other animals are also having at least visually with a total secondary question of whether it's experiential or not but just visually so active inference is a process theory that we can go into more another time that has to do with how organisms actually act in their ecological niche to resolve their uncertainty about the kinds of problems they actually need to resolve therefore framing things like foraging as a reward obtaining but a precision maximization behavior sequence and that framing turns out to have big implications so just wanted to frame it that way as far as the scientist's active the scientist also from an informational has to think about which hypotheses experimenting or modeling with to reduce their uncertainty about what and just as in active inference where the generative model of the internal state the organism in this example the scientist the internal model includes a generative model of the latent causes in the world and understanding the latent causes better is what facilitates uh better action and also better experiments and hopefully clearer transmission as well how does the markov blanket idea come into play what pierces the blanket so here any figure that you look at of the markov blanket is going to have uh basically the active inference cycle with a markov blanket superimposed down the middle there's internal and external states and then external states influence sensory states sensory states influence internal states internal states have their own autonomous or self-influencing dynamics and through policy selection end up implementing action states for example the movement of an arm action states influence the external states the world states for example which are also of course involved in their own dynamics the markov blanket is pierced on one side by sense coming in and by action on the other side going out and then the whole multi-scale question is going to be about applying that markov blanket all the way down idea taking that seriously as far as cognition one other area that i

thought could be kind of interesting to relate this to would be like other partitioning problems and another partitioning problem that comes up in biological philosophy realms is this question of nature versus nurture and here i would point people towards two pretty recent and excellent books on the left is beyond versus the struggle to understand the interaction of nature and nurture by james tabary and then on the right the mirage of a space between nature and nurture by evelyn fox keller and they both have a different perspective on the same question but in both cases there's this idea of moving beyond the binary partitioning of nature and nurture and so here in the ramstead 2019 paper the framing of internalism and externalism where they're sort of entering into this stack is at the level of cognition it's about cognitive entities whereas here's a partitioning problem in the area of genetics and in environmental and developmental studies for example and here's an example of where similar internalism internals the genes right smaller things or externalism outside bigger things the ecosystem is seen as bigger than the person with this implicit focused level of analysis on the individual and so um just to draw on evelyn fox keller's book she uses a metaphor of different kinds of tasks and how even if you just had two people so the same type of agent if you had two people building a brick wall together you could at the end of the day say well this person added so and so many bricks and this person added the rest of the bricks and so maybe you could say well 40 of the work on this wall was performed by that person however imagine a situation where two people are needed to hold a hose to fill up a bucket for example even if one of the rules requires a specialty or more training or some other attribute even in that case that person can't do it alone so no matter how much water comes through that hose and it's all water coming through the hose you're never going to be able to say well 80 of the water was due to person a and 20 percent was due to person b so she uses that really clear metaphor to describe the relationship between nature and nurture how they're obviously related there is no such thing as a uh niche free organism especially when you think about this from an evolutionary ecology or ultimately active inference organisms and their niche are what fitness is about that is what is being fit to what is the organism to their niche and you don't see things that aren't fit so it's unsurprising that we're finally attuned to things in our niche also there was a one interesting section in mirage of a space between where she traces the linguistic origins of the framing nature versus nurture because everyone has heard the phrase nature versus nurture is it nature or is it nurture or it's in our dna it's in our nature how did things come to be in this verses sense that's also alluded to in the of tabari's book and what fox keller writes is recently a fashion has arisen for tracing the phrase nature and nurture and the debate with which that

phrase is associated back to shakespeare or at least to prospero in the tempest 1623 who writes off caliban as uneducable quoting shakespeare a born devil on whose nature slash nurture will never stick back to fox killer some have traced it further back to a monograph on children's education written by an elizabethan pedagogue richard mulcaster mulcaster's words were nature makes boy towered nurture sees him forward maybe it rhymes in another pronunciation it's from 1581 and then she concludes but in fact although both molecaster and shakespeare juxtapose the workings of nature and nurture neither invokes the term as an explicit conjunction so not an or they do not write of nature and nurture as such so they don't separate them and they don't um make them contra distincted versus each other indeed when we look more closely at what they do right we can see that their use of the two terms does not invite such a conjunction for there is no presumption of an a priori disjunction so what she's getting at there is there really is no need to do an argument against the concept of nature and nurture it's kind of like writing a hit article on somebody who you disagree with people who understand that there has to be a move beyond versus beyond an oppositional framing of nature and nurture those people should be and are making their positive arguments for why it should be framed a different way where the nature and the nurture are things that are pulled out or controlled for specifically and transparently in specific systems so that's an example about how the philosophy of science for example how we think about the partition between internal and external states of phenotype relates to how we think about multi-scale systems overall because i think okay it reminds me so much of this internalism and externalism in cognitive sciences debate reminds me a lot of this historical debate in genetics all right next section perspectives on systems and action and here the motivating question is about how do we carve nature at the joints which is a quote attributed to plato in the phaedrus allegedly and the quote alludes to this idea that nature whether one thinks that systems can be defined tightly and they're just on their own billiard ball trajectory or whether one has a relational perspective where there's just these multi-level networks of relations and being and becoming whatever it is at some point we want to carve it at the joints because otherwise we're going to get swept up in the storm and people will often hold that that's not their position or that they don't want to carve systems at the joints but i think if you're listening to this i hope you can see that at some point at least in action you have to be dealing with nature at the joints so to speak are they the right ones or not isn't that the whole question so one thing that is a connection is we can ask how is the fep the free energy principle or active inference going to help us frame this problem of carving nature at the joints this is kind of one of the big ideas from a philosophical

perspective is about classification and about distinguishing things whether it's objects or types of processes or um ideas distinguishing things and those are part of nature is part of philosophy and so coming back to this essential question of carving nature at the joint and then asking well how can we use things like fep active inference to frame the problem structurally to give us in ontology and in fact in the paper they'll talk about an ontology subsequent work also um worked on ontologies so that's sort of the big motivation it's why this quote is being brought up in the context of this paper we're about to discuss because the whole point is about us carving nature at the joints so let's go a little bit deeper into systems modeling and carving nature at the joints and the question is how are descriptive we'll just call that type 1 and action oriented type 2 models related or different and so just to explain what those 1 and 2 are by descriptive model i mean a system description only and so two examples just to sort of pin down the qualitative to quantitative spectrum the first mention here is to bruno latour's actor network theory or ant and it's an awesome piece of maybe meta sociology not sure exactly what the torah would call it that just says the way to approach the system is by showing up finding out what are the traces and being open with respect to which actors are involved in which processes through which networks of influence so it's a very unbiased description or moving towards unbiased not to step into that sort of uh trap i guess and then um it's very qualitative it's relational it's also about the observers knowing and relationship with the system not just the way that aspects of the system are associated with each other so often those things are left out in quantitative or in some scientific world use and on the other end of the system's description only spectrum i've pointed to a package of complex systems design tools called cadcad now cad cad goes beyond systems description it turns out to also be able to do things like simulation and and other more dynamical approaches like signal processing but what i mean by putting cad cad here is that it can be used purely depictively to specify a complex especially one that's computational so there might be a question when you're actually trying to model a organic ecosystem of how do you measure some every single uh you know piece of the ecosystem and how many variables is enough to describe the fineness of the temperature gradient or something like that but when the simulation is about a digital system it can be more deterministically and controllably simulated if one can imagine so there's one type of world modeling related to system description but that obviously enables simulation and inference as well the second kind of models are action-oriented models so action-oriented models are those where ultimately we're just we're assigning meaning to states voting with your fee voting with your dollars it's when we want to have effective action or at

least the system acts like it wants to something like maybe what denett would call the intentional stance so the descriptive modeling might be a question that you're asking the computer like given this scatter of points which axes capture the most statistical variants and that would be just it running some sort of algorithm to find the maximum variance explained whereas a another type of algorithm like more of a type 2 action-oriented algorithm might be learn the relationship between this input and the optimal performing behavior in the output so that's what action-oriented models are about next question how does the multi-level perspective influence our answers above so if we're going to have a story where there's nature's being carved at its joints maybe around the markov blankets maybe with scents coming in and action coming out even if we have that at one level so we have an asian-based model of the ants that are bumping into each other and at one level the sense of one is the action of another and so we have a sort of bumper car model um where does the multi-level perspective come into play and that's kind of thinking about the bumper car one it's where you would model first the cars bumping into each other and then you could go into the engine or you could go out to the power grid and to the people who are riding on the bumper cars or something like that so how are we going to deal with the complexity of causal stories relationships some of what might which uh be really important and like the example of a molecule that binds to a receptor others which might be spurious and the reality is that because of the combinatorics most of the options across levels will be spurious so most of the ecological relationships can't be for example one-to-one because there's so many that are just bumping into each other all the time just to give one example from ecology so how are we going to carve nature at its joints do it cleanly are we trying to represent stuff like one are we trying to act like do one and two fall out as a function of doing one do we get one and two as a function of following out just two how does emergence and emergent properties fit into all of this so by emergence or emergent properties people often mean a sort of system outcome where there's some property or characteristic or attribute that is arising at a higher level again bigger broader spatial and temporal scales based upon interactions or some sort of interfacing of the smaller level so sometimes these emergent properties using the classification here of badao are in the weak category and in the strong category it's not the only way to break down immersions not to only take here but i find this one pretty helpful first strong emergence badao equates with basically magic you can read the papers more if you want to explore more nuance but the strong emergence is from something that's totally disjoint to something that's has a total integrity like language or awareness for example and then weak emergence is contrasted

and that's most of the physical systems where even if the relationships between subunits are really nuanced or the computational uh capacity required to simulate the system would be extreme those systems are still weakly emerging in that it still just is ants bumping into each other now they're developing and that's sort of crazy and evolution shapes that and everything but still we're talking about a physical system not in the genre for example of um awareness arising spontaneously so how is that going to play into uh the discussion with active inference showing up on the scene where will this relate to immersions these are things anyone's welcome to chime in on also we can tie the idea of emergence and the multi-scale systems approach to this debate that we often hear about reductionism versus holism so reductionism is simply a method of explanation where bigger things are made sense of by appeals to the properties of their smaller things like the bigger wall is um orange because each of the smaller pixels is orange or something like that there's a variety of reductionist explanations and it's just one of our hands it's one of our approaches or our wings holism is kind of flipping that uh schema of reductionism saying actually smaller things are made sensitive in terms of the bigger thing and that um is kind of the opposite edge viewpoint and so we have some sliding to toward a holistic understanding some sliding towards a reductionist understanding a lot of philosophical work in this area and the thing that we would just want to note here would just be first ask continually how is the free relating to reductionism versus holism because sometimes it may feel like there is a sort of a philosophy of science shift associated with taking on the active inference framework and so what are the implications of taking that seriously where does that put this reductionism versus holism is it going to be put into a similar bin as nature versus nurture as kind of a turn of phrase or a false dichotomy that was just eventually seen through or will there be people who still think that that sort of binary phrasing whichever side they find themselves on is going to be appropriate and again this is where dennis noble's work on the bottom up and top down is related the other work here is girdle escherbach which is the 1979 book by douglas hofstetter and here's one of the figures from the book where it says uh mu mu i don't know how you'd say it which is a allegorically at least a sort of innovative response or a canonical response depending on who you ask of zen koan and zen is a topic that's brought up in the book and the m is written of letters that are containing smaller letters that say reductionism and then the u is made up of letters spell out the other words so it's kind of this multi-scale um thought loop really captures the spirit of the book well and also it captures at least informally through art as often is the case what it looks like to go beyond the reductionism versus holism debate so first if only in a text illustration

but then eventually how does that relate to a computer screen how about a person interacting with a computer screen how about two people interacting with their computer screens all right few more perspectives on systems modeling and action how is world modeling related to pragmatics and action so the questions to ask here is really is there a best map of the territory uh short answer no and this is true for geospatial maps as well as fields of affordances which we talked about in the previous active streams which also often demand to be mapped for example skill set mappings or concept mappings so there's a lot to say about map and territory of course but for here it's suffices to say that the best map is defined by the mission by the context by the degree of certainty by all these other features and factors related to the user and the situation so we're talking about making maps uh of meaning maps for the world we're talking about doing world modeling for effective inference and action as we were talking about previously so here's just one quote from the paper motivate you towards the things that you might be able to hear them expand more on they write the multi-scale free energy principle model allows us to cast the boundaries of cognition as assembled and maintained in an informational dynamics across multiple spatial and temporal scales crucially we shall show that this multi-scale application of the fep free energy principle implies both ontological and methodological pluralism so a few thought questions whether this is before or after reading the paper you could ask is the free energy principle pluralistic with respect to non free energy principle theories or frameworks or principles which levels or types of theory is the fep consistent or discordant with and one starting point there would be searching for carl fristen's interview in the 2018 alias bulletin and it's an interesting question to ask about any theory is okay well who is this changing the mind of or what is the opposite of this theory and is it true or does anyone hold the true opposite of this theory does anyone hold a 50 opposite is it 50 by outcome or is it fifty percent by axiom in that they diverge so you can always ask that and i found that that was a very um interesting route to go down as far as the fep another quote from the paper is just because it's such a direct claim our claim is that which among these which are the situated barriers of talking of systems which among these are the most relevant will depend on the phenomena being studied and the explanatory interests of researchers and so there's a few notions like markov blanket which we talked about earlier is that sense in action piercing that's related to ecological psychological concepts as well the markov blankets are defined in a few different ways they're defined by statistical covariations but one type of those is going to be causal and also another type of those could be like a measurement and so the kinds of measurements that we make the types of causal relationships that are important and

then which models we apply to them is going to depend on uh what is enacted by the researchers because they're also part of embodied and cultured cognition so briefly what are mythological and ontological pluralism so methodological pluralism means using a variety of methods or ways of doing ontological pluralism means having a variety of ontologies or ways of knowing and pluralism is referring to there being multiple now there are other schools of thought that are sort of higher order than pluralism as to how do you cut through that is it just theory anarchy like um fair bend suggested where the pluralism can just be taken to this extreme uh equivalent to almost a novelty search in theory space or is there going to be some way to cut through that and determine which of the pluralistic options or possibilities or actualities is preferable because ultimately we do have to act that's that type 2 type of model so ask yourself for direct action or for more philosophical topics what is utilitarianism or pragmatism how do such ideas relate to pluralism so it's one thing to say we're scientists and whichever level of analysis is most relevant well that's going to depend on what we're studying and what we're interested in explaining but should it is that really the best way to approach these systems should we really um knowing that we can never remove for example individual bias but still wanting to model systems so that we can act effectively how are we going to balance just this sort of ruthless utilitarianism pragmatism whatever ends justify the means um whatever preconceived worldview whatever your bayesian prior is just stick with that no matter what happens how how is that related to cutting through the noise of pluralism these are again just questions to raise in active inference how are ideas related to action and what kind of ideas or mental states are there there's a few kind of interesting categories that we perceive at least experientially various schools of thought on that but where are we going to fit these into the active inference framework are those just different loops in our generative model things like emotion or desires or dreams or secrets where are those going to fit into this active inference framework how integrated is that going to be across types of ideas and how are our conserved or novel ideas respectively meaning orthodoxy or heterodoxy associated with the expression of conserved or novel actions so that is ortho practice or heteropraxis and linking that to active inference which is kind of just framing this two-stroke engine or this continual ooda loop a lot of ways to think about it it's just framing this continual flow between action and inference so how are those two linked here's another way of asking and just to conclude with two books to study how this has been studied from a philosophy of science perspective on the right is scientific pluralism uh edited by keller longino and waters and then on the left is a book called studying human behavior how scientists investigate aggression and sexuality by

helen longneau and these books talk about the question of pluralism in science and different ways that people have sort of cut through the massive theories or contributed to it i guess in some cases and there's just a really interesting body of thought and so it's uh useful to think for people who are in the free energy space as well as from the outside what are the bridges that link what's happening with the free energy principle and active inference to more classical philosophy of science and so hopefully in the first in alias paper from 2018 and also some other excellent work from the last few years we'll come to a better understanding of that so thanks for participating which i guess was me and you as the listener we will provide a follow-up form to live participants and we would welcome any feedback suggestions questions you have and yeah stay in communication with us so thanks again for listening and we hope to either see you at 5.1 or a future discussion